Math for EE

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About this Book

This book introduces matrices through high school coordinate geometry. This approach makes it easier for beginners to learn Python for scientific computing. All problems in the book are from NCERT mathematics textbooks from Class 9-12. Exercises are from CBSE board exam papers.

The content is sufficient for industry jobs and covers nearly all matrix prerequisites for machine learning. There is no copyright, so readers are free to print and share.

First manual appeared in January 2018
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In this edition, some incorrect solutions were removed. All figures redrawn. More problems added.

Contents

1. GATE AE 2026

A. *Formulae*

1.1.1 Probability

- Addition Theorem of Probability:

$$P(M) + P(E) - P(ME) + P\left((M \cup E)'\right) = 1 \tag{1.1.1.1}$$

- Complement Laws in Boolean algebra

$$A + A' = 1, \quad AA' = 0 \tag{1.1.1.2}$$

- Probability of Certain and Impossible Events

$$P(1) = 1, \quad P(0) = 0 \implies P(A + A') = 1, \quad P(AA') = 0 \tag{1.1.1.3}$$

- Addition Law of Probability for Mutually Exclusive Events

$$P(A + A') = P(A) + P(A') \tag{1.1.1.4}$$

- Difference rule of probability

$$P(AB') = P(A) - P(AB) \tag{1.1.1.5}$$

Proof:
Using (1.1.1.2)

$$A = A \cdot 1 = A(B + B') \tag{1.1.1.6}$$

Applying the distributive law:

$$A = AB + AB' \tag{1.1.1.7}$$

To apply the additive property of probability, we check the intersection of AB and AB':

$$(AB)(AB') = A \cdot A \cdot B \cdot B' = A \cdot (BB') \tag{1.1.1.8}$$

Since $BB' = 0$:

$$(AB)(AB') = A \cdot 0 = 0 \tag{1.1.1.9}$$

Thus, the events AB and AB' are mutually exclusive.

Since $A = AB + AB'$ and $(AB)(AB') = 0$, from (1.1.1.4)

$$P(A) = P(AB + AB') = P(AB) + P(AB') \tag{1.1.1.10}$$

Rearranging the equation (3.9) to isolate $P(AB')$:

$$P(AB') = P(A) - P(AB) \tag{1.1.1.11}$$

1.1.2 Laplace Transform

- Laplace transform on $g'(t)$

  Using the definition of the Laplace transform:

$$\mathcal{L}\{g'(t)\} = \int_0^\infty e^{-st} g'(t)\ dt \tag{1.1.2.1}$$

  Applying integration by parts with $u = e^{-st}$ and $dv = g'(t)\ dt$ (giving $du = -se^{-st}\,dt$ and $v = g(t)$):

$$\mathcal{L}\{g'(t)\} = \left[e^{-st} g(t)\right]_0^\infty - \int_0^\infty \left(-se^{-st}\right) g(t)\ dt \tag{1.1.2.2}$$

$$= \left(\lim_{t\to\infty} e^{-st} g(t) - g(0)\right) + s\int_0^\infty e^{-st} g(t)\ dt \tag{1.1.2.3}$$

$$\mathcal{L}\{g'(t)\} = 0 - g(0) + sG(s) = sG(s) - g(0) \tag{1.1.2.4}$$

- Laplace transform on $g''(t)$

  Using the definition for $g''(t)$:

$$\mathcal{L}\{g''(t)\} = \int_0^\infty e^{-st} g''(t)\ dt \tag{1.1.2.5}$$

  Applying integration by parts with $u = e^{-st}$ and $dv = g''(t)\ dt$ (giving $du = -se^{-st}\,dt$ and $v = g'(t)$):

$$\mathcal{L}\{g''(t)\} = \left[e^{-st} g'(t)\right]_0^\infty - \int_0^\infty \left(-se^{-st}\right) g'(t)\ dt \tag{1.1.2.6}$$

$$= \left(\lim_{t\to\infty} e^{-st} g'(t) - g'(0)\right) + s\int_0^\infty e^{-st} g'(t)\ dt \tag{1.1.2.7}$$

  With $g'(0) = 0$ and substituting the result from part (a), $\mathcal{L}\{g'(t)\} = sG(s) - g(0)$:

$$\mathcal{L}\{g''(t)\} = 0 - g'(0) + s\left(sG(s) - g(0)\right) = s^2 G(s) - sg(0) - g'(0) \tag{1.1.2.8}$$

- The First Frequency Shift Theorem: It states that if $\mathcal{L}\{f(x)\} = F(s)$, then:

$$\mathcal{L}^{-1}\{F(s-a)\} = e^{ax} f(x) \tag{1.1.2.9}$$

  Proof:

  By definition, the Laplace transform of a function $g(x)$ is:

$$\mathcal{L}\{g(x)\} = \int_0^\infty e^{-sx} g(x)\ dx \tag{1.1.2.10}$$

  Substituting $g(x) = e^{ax} f(x)$:

$$\mathcal{L}\{e^{ax} f(x)\} = \int_0^\infty e^{-sx} \left(e^{ax} f(x)\right) dx \tag{1.1.2.11}$$

$$= \mathcal{L}\{e^{ax} f(x)\} = \int_0^\infty e^{-(s-a)x} f(x)\, dx \tag{1.1.2.12}$$

Recall that $F(s)$ is defined as:

$$F(s) = \int_0^\infty e^{-sx} f(x)\, dx \tag{1.1.2.13}$$

Replacing s with $(s-a)$:

$$F(s-a) = \int_0^\infty e^{-(s-a)x} f(x)\, dx \tag{1.1.2.14}$$

Therefore:

$$\mathcal{L}\{e^{ax} f(x)\} = F(s-a) \tag{1.1.2.15}$$

Applying the inverse Laplace operator $\mathcal{L}^{-1}$ to both sides yields:

$$\mathcal{L}^{-1}\{F(s-a)\} = e^{ax} f(x) \tag{1.1.2.16}$$

- Inverse Laplace Transform on $1/s^2$:

$$\mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\} = x \tag{1.1.2.17}$$

Proof:

$$\mathcal{L}(x) = \int_0^\infty x e^{-sx}\, dx \tag{1.1.2.18}$$

$$= \left(-\frac{x}{s} e^{-sx}\right)_0^\infty + \frac{1}{s}\int_0^\infty e^{-sx}\, dx \tag{1.1.2.19}$$

$$= 0 + \frac{1}{s}\left(-\frac{1}{s} e^{-sx}\right)_0^\infty = \frac{1}{s^2} \tag{1.1.2.20}$$

$$\implies \mathcal{L}^{-1}\left(\frac{1}{s^2}\right) = x \tag{1.1.2.21}$$

- Inverse Laplace Transform to produce the cosine and sine functions

$$\mathcal{L}\left(e^{iax}\right) = \int_0^\infty e^{-sx} e^{iax}\, dx = \int_0^\infty e^{-(s-ia)x}\, dx \tag{1.1.2.22}$$

$$= \left(-\frac{e^{-(s-ia)x}}{s-ia}\right)_0^\infty = \frac{1}{s-ia} \tag{1.1.2.23}$$

$$= \frac{s+ia}{(s-ia)(s+ia)} = \frac{s}{s^2+a^2} + i\frac{a}{s^2+a^2} \tag{1.1.2.24}$$

We know that

$$\mathcal{L}\left(\cos(ax) + i\sin(ax)\right) = \mathcal{L}\left(\cos(ax)\right) + i\,\mathcal{L}\left(\sin(ax)\right) : \tag{1.1.2.25}$$

Equating the real and imaginary parts, we get:

$$\mathcal{L}\left(\cos(ax)\right) = \frac{s}{s^2+a^2} \implies \mathcal{L}^{-1}\left(\frac{s}{s^2+a^2}\right) = \cos(ax) \tag{1.1.2.26}$$

$$\mathcal{L}\left(\sin(ax)\right) = \frac{a}{s^2+a^2} \implies \mathcal{L}^{-1}\left(\frac{a}{s^2+a^2}\right) = \sin(ax) \tag{1.1.2.27}$$

- Solving a differential equation of the form:

$$\frac{d^2y}{dx^2} = -\alpha^2 y, \alpha^2 \in \mathbb{R}^+ \tag{1.1.2.28}$$

Applying the Laplace Transform on both sides, and using (1.1.2.8), we get:

$$s^2G(s) - sy(0) - y'(0) = -\alpha^2 G(s) \tag{1.1.2.29}$$

Let $y(0) = \beta$ and $y'(0) = \gamma$
Rearranging, we get:

$$G(s) = \frac{s\beta + \gamma}{s^2 + \alpha^2} \tag{1.1.2.30}$$

$$G(s) = \frac{s\beta}{s^2+\alpha^2} + \frac{\gamma}{s^2+\alpha^2} \tag{1.1.2.31}$$

From results (1.1.2.26) and (1.1.2.27), we get:

$$G(s) = \beta cos(\alpha x) + \frac{\gamma}{\alpha} sin(\alpha x) \tag{1.1.2.32}$$

1.1.3 Euler's Method of Forward Differences

- First Forward Difference:
Consider a differential equation:

$$\frac{dy}{dx} = f\left(x, y\right), y\left(x_0\right) = y_0 \tag{1.1.3.1}$$

The slope of the tangent m at $x = x_0$ is given by:

$$m = f\left(x_0, y_0\right) \tag{1.1.3.2}$$

The slope of the tangent line gives the direction in which we need to move in order to trace the curve(considering we move in small steps). Hence, we can numerically compute the approximate value of a function at a given x knowing the initial conditions.

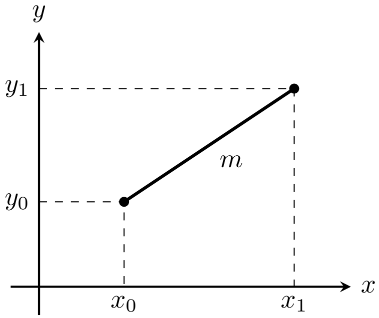

Fig. 1.1.3.1 Zoomed in graph

Consider the above diagram, the equation of line is:

$$y_1 - y_0 = m\,(x_1 - x_0) \tag{1.1.3.3}$$

$$y_1 = y_0 + mh \tag{1.1.3.4}$$

Here, $h = x_1 - x_0$ is the step size(or small increment in x)
Generalising the above result, we get the derivative y_n':

$$y_n' = \frac{y_{n+1} - y_n}{h} \tag{1.1.3.5}$$

- Second Forward Difference:

Differentiating the equation (1.1.3.5), we obtain:

$$y_n'' = \frac{y_{n+1}' - y_n'}{h} \tag{1.1.3.6}$$

From equation (1.1.3.5), we obtain:

$$y_{n+1}' = \frac{y_{n+2} - y_{n+1}}{h}, y_n' = \frac{y_{n+1} - y_n}{h} \tag{1.1.3.7}$$

Substituting, we get:

$$y_n'' = \frac{y_{n+2} - 2y_{n+1} + y_n}{h^2} \tag{1.1.3.8}$$

1.1.4 The Fourier Series

Consider a function $f(t)$ that is periodic. This function can be expressed as a sum of sines and cosines as shown below. This sum is called the Fourier series.

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(n\omega_0 t) + \sum_{n=1}^{\infty} b_n \sin(n\omega_0 t) \tag{1.1.4.1}$$

To find the unknown coefficients a_0, a_n, and b_n, we use the orthogonality properties of trigonometric functions over one full period T.

Orthogonality Relationships:

For any non-zero integers m and n evaluated over one period $\int_0^T dt$:

- $\int_0^T \cos(m\omega_0 t)\,dt = 0$
- $\int_0^T \sin(m\omega_0 t)\,dt = 0$
- $\int_0^T \cos(m\omega_0 t)\sin(n\omega_0 t)\,dt = 0$
- $\int_0^T \cos(m\omega_0 t)\cos(n\omega_0 t)\,dt = \begin{cases} 0, & m \neq n \\ \frac{T}{2}, & m = n \neq 0 \end{cases}$
- $\int_0^T \sin(m\omega_0 t)\sin(n\omega_0 t)\,dt = \begin{cases} 0, & m \neq n \\ \frac{T}{2}, & m = n \neq 0 \end{cases}$

Deriving a_0 (DC Component):
Integrate both sides of the expansion equation over one period $[0, T]$:

$$\int_0^T f(t)\,dt = \int_0^T \frac{a_0}{2}\,dt + \sum_{n=1}^{\infty} a_n \int_0^T \cos(n\omega_0 t)\,dt + \sum_{n=1}^{\infty} b_n \int_0^T \sin(n\omega_0 t)\,dt \tag{1.1.4.2}$$

By orthogonality, every integral inside the summations equals zero:

$$\int_0^T f(t)\,dt = \frac{a_0}{2}T \implies a_0 = \frac{2}{T}\int_0^T f(t)\,dt \tag{1.1.4.3}$$

Deriving a_n (Cosine Coefficients):

Multiply both sides of the expansion equation by $\cos(m\omega_0 t)$ and integrate over $[0, T]$:

$$\begin{aligned} \int_0^T f(t)\cos(m\omega_0 t)\,dt = {} & \frac{a_0}{2}\int_0^T \cos(m\omega_0 t)\,dt \\ & + \sum_{n=1}^{\infty} a_n \int_0^T \cos(n\omega_0 t)\cos(m\omega_0 t)\,dt \\ & + \sum_{n=1}^{\infty} b_n \int_0^T \sin(n\omega_0 t)\cos(m\omega_0 t)\,dt \end{aligned} \tag{1.1.4.4}$$

Applying orthogonality:

- The a_0 integral equals 0.
- The b_n integrals equal 0 for all n.
- The a_n integrals equal 0 for every term except where $n = m$:

$$\int_0^T f(t)\cos(m\omega_0 t)\,dt = a_m \int_0^T \cos^2(m\omega_0 t)\,dt = a_m \left(\frac{T}{2}\right) \tag{1.1.4.5}$$

Replacing dummy index m with n:

$$a_n = \frac{2}{T}\int_0^T f(t)\cos(n\omega_0 t)\,dt \tag{1.1.4.6}$$

Deriving b_n (Sine Coefficients)

Multiply both sides of the expansion equation by $\sin(m\omega_0 t)$ and integrate over $[0,T]$:

$$\begin{aligned}\int_0^T f(t)\sin(m\omega_0 t)\,dt = {} & \frac{a_0}{2}\int_0^T \sin(m\omega_0 t)\,dt \\ & + \sum_{n=1}^{\infty} a_n \int_0^T \cos(n\omega_0 t)\sin(m\omega_0 t)\,dt \\ & + \sum_{n=1}^{\infty} b_n \int_0^T \sin(n\omega_0 t)\sin(m\omega_0 t)\,dt \end{aligned} \tag{1.1.4.7}$$

Applying orthogonality:

- The a_0 integral equals 0.
- The a_n integrals equal 0 for all n.
- The b_n integrals equal 0 for every term except where $n = m$:

$$\int_0^T f(t)\sin(m\omega_0 t)\,dt = b_m \int_0^T \sin^2(m\omega_0 t)\,dt = b_m\left(\frac{T}{2}\right) \tag{1.1.4.8}$$

Replacing dummy index m with n:

$$b_n = \frac{2}{T}\int_0^T f(t)\sin(n\omega_0 t)\,dt \tag{1.1.4.9}$$

1.1.5 Matrices

- Fundamental property:

$$\left|\mathbf{A}^{\mathbf{T}}\right| = \left|\mathbf{A}\right| \tag{1.1.5.1}$$

Proof:
When expanding $\left|\mathbf{A}\right|$ along Row 1, you select entries across the top row and compute the determinants of their remaining submatrices.
When expanding $\left|\mathbf{A}^{\mathbf{T}}\right|$ down Column 1, those exact same entries now sit vertically in the first column, and their corresponding submatrices are simply transposed.
Visual Matrix Comparison:
Matrix $\mathbf{A}$ (Expand along Row 1):

$$\left|\mathbf{A}\right| = \begin{vmatrix} \mathbf{a_{11}} & \mathbf{a_{12}} & \mathbf{a_{13}} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} \tag{1.1.5.2}$$

Matrix $\mathbf{A^T}$ (Expand down Column 1):

$$\left|\mathbf{A^T}\right| = \begin{vmatrix} \mathbf{a_{11}} & a_{21} & a_{31} \\ \mathbf{a_{12}} & a_{22} & a_{32} \\ \mathbf{a_{13}} & a_{23} & a_{33} \end{vmatrix} \tag{1.1.5.3}$$

Submatrix Comparison for Entry a_{12}:

When selecting a_{12}:

- In $\mathbf{A}$ (Row 1, Column 2 deletion):

$$\mathbf{M_{12}} = \begin{pmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{pmatrix} \tag{1.1.5.4}$$

- In $\mathbf{A^T}$ (Row 2, Column 1 deletion):

$$\mathbf{N_{21}} = \begin{pmatrix} a_{21} & a_{31} \\ a_{23} & a_{33} \end{pmatrix} \tag{1.1.5.5}$$

Notice that $\mathbf{N_{21}} = (\mathbf{M_{12}})^T$. Since $\left|\mathbf{M_{12}}\right| = \left|(\mathbf{M_{12}})^T\right|$, every term in the expansion of $\left|\mathbf{A}\right|$ is identically equal to the corresponding term in $\left|\mathbf{A^T}\right|$.

- A few relations on the eigenvalues of a matrix
  Let $\lambda_1, \lambda_2, ..., \lambda_i$ be the eigenvalues of a matrix $\mathbf{A}$. Then, we have:

$$\sum_{i=1}^{n} \lambda_i = \text{Tr}\left(\mathbf{A}\right) \tag{1.1.5.6}$$

$$\prod_{i=1}^{n} \lambda_i = \left|\mathbf{A}\right| \tag{1.1.5.7}$$

- Determining the type of conic
  Any conic section in a 2D plane (p, q) is expressed by the general equation:

$$S\left(p, q\right) = ap^2 + 2hpq + bq^2 + 2gp + 2fq + c = 0 \tag{1.1.5.8}$$

Given the general second degree equation of a conic, the determinant of the matrix $A = \begin{pmatrix} a & h \\ h & b \end{pmatrix}$ determines what type of conic we are dealing with.

- If $\left|A\right| > 0$, the conic is an Ellipse
- If $\left|A\right| = 0$, the conic is a Parabola
- If $\left|A\right| < 0$, the conic is a Hyperbola

Proof:
Let's consider the case where $\left|A\right| < 0$. From (1.1.5.7), because $\left|A\right| < 0$, the two eigenvalues must have opposite signs—one strictly positive $(\lambda_1 > 0)$ and one strictly negative $(\lambda_2 < 0)$. A quadratic form whose matrix has eigenvalues of mixed signs is defined as indefinite (it can take both positive and negative values depending on the direction vector).

By rotating the coordinate axes along the principal eigenvectors of A, the cross-term $2hxy$ vanishes, simplifying the second-degree terms to:

$$\lambda_1 X^2 + \lambda_2 Y^2 \tag{1.1.5.9}$$

Substituting $\lambda_2 = -|\lambda_2|$ (since $\lambda_2 < 0$) yields:

$$\lambda_1 X^2 - |\lambda_2| Y^2 + \text{linear terms} = C \tag{1.1.5.10}$$

Completing the square and shifting the origin brings the equation into the familiar standard form of a hyperbola:

$$\frac{(X - X_0)^2}{\alpha^2} - \frac{(Y - Y_0)^2}{\beta^2} = 1 \tag{1.1.5.11}$$

Similar proofs can be written for the other conics.

B. Problems

1.2.1 How many 3-digit numbers can be formed using three distinct single digit prime numbers? (GATE AE 2026)

a) 64
b) 24
c) 12
d) 4

Solution: There are 4 single digit prime numbers: 2,3,5 and 7. The desired numbers are given by

$$^4P_3 = \frac{4!}{1!} = 24 \tag{1.2.1.1}$$

1.2.2 In a group of students, 10 students like Mathematics, 12 students like English, 4 students like both Mathematics and English, and 6 students like neither Mathematics nor English. The number of students in the group is ______. (GATE AE 2026)

a) 18
b) 20
c) 24
d) 32

Solution:
Let N be the total number of students in the group.
If one student is chosen uniformly at random from the group, define:

- M: The event that the student likes Mathematics.
- E: The event that the student likes English.

The respective probabilities are expressed as:

$$P\left(M\right) = \frac{10}{N}, \quad P\left(E\right) = \frac{12}{N} \tag{1.2.2.1}$$

$$P\left(ME\right) = \frac{4}{N} \quad \text{(student likes both)} \tag{1.2.2.2}$$

$$P\left((M+E)'\right) = \frac{6}{N} \quad \text{(student likes neither)} \tag{1.2.2.3}$$

Since the total probability of the sample space is 1:

$$P\left(M \cup E\right) + P\left((M+E)'\right) = 1 \tag{1.2.2.4}$$

From (1.1.1.1)

$$P\left(M\right) + P\left(E\right) - P\left(ME\right) + P\left((M+E)'\right) = 1 \tag{1.2.2.5}$$

Substitute the probabilities in terms of N:

$$\frac{10}{N} + \frac{12}{N} - \frac{4}{N} + \frac{6}{N} = 1 \tag{1.2.2.6}$$

$$\frac{10+12-4+6}{N} = 1 \tag{1.2.2.7}$$

$$\frac{24}{N} = 1 \implies N = 24 \tag{1.2.2.8}$$

1.2.3 Given a square of side $s = 4$cm and two circles, one with known radius $r_1 = 1$cm and another with unknown radius r_2, such that both of them touch each other externally and also the square, find the value of r_2. Also describe how the figure in the problem can be constructed. (GATE AE 2026)

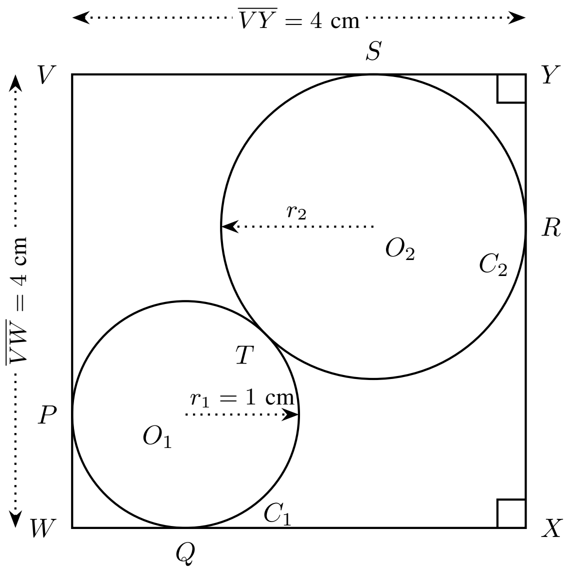

Fig. 1.2.3.1 Touching Circles and Square

a) **Solution:**
Let the side of the square be s, radius of the first circle be r_1 and that of the unknown circle be r_2.

Clearly from figure:

$$WO_1 + O_1T + TO_2 + O_2Y = \sqrt{2}s \tag{1.2.3.1}$$

We know that:

$$WO_1 = \sqrt{2}r_1, O_1T = r_1, TO_2 = r_2, O_2W = \sqrt{2}r_2. \tag{1.2.3.2}$$

Substituting this in our result from point (4.2) we get:

$$r_2 = \frac{\sqrt{2}s - \left(\sqrt{2}+1\right) r_1}{\sqrt{2}+1}. \tag{1.2.3.3}$$

Substituting the values of s and r_1, we get:

$$r_2 = \left(7 - 4\sqrt{2}\right) \text{cm} \tag{1.2.3.4}$$

b) **General method for construction:**
Vertices of the square:

$$\mathbf{W} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad \mathbf{X} = \begin{pmatrix} s \\ 0 \end{pmatrix}, \quad \mathbf{Y} = \begin{pmatrix} s \\ s \end{pmatrix}, \quad \mathbf{V} = \begin{pmatrix} 0 \\ s \end{pmatrix} \tag{1.2.3.5}$$

Circle C_1 (r_1):

$$\mathbf{O}_1 = \begin{pmatrix} r_1 \\ r_1 \end{pmatrix}, \quad \mathbf{P} = \begin{pmatrix} 0 \\ r_1 \end{pmatrix}, \quad \mathbf{Q} = \begin{pmatrix} r_1 \\ 0 \end{pmatrix} \tag{1.2.3.6}$$

Circle C_2 (r_2):

$$\mathbf{O}_2 = \begin{pmatrix} s - r_2 \\ s - r_2 \end{pmatrix}, \quad \mathbf{S} = \begin{pmatrix} s - r_2 \\ s \end{pmatrix}, \quad \mathbf{R} = \begin{pmatrix} s \\ s - r_2 \end{pmatrix} \tag{1.2.3.7}$$

Line of centers and contact point $\mathbf{T}$:

$$\mathbf{O}_2 - \mathbf{O}_1 = \begin{pmatrix} s - r_2 - r_1 \\ s - r_2 - r_1 \end{pmatrix} = (s - r_2 - r_1) \begin{pmatrix} 1 \\ 1 \end{pmatrix} \tag{1.2.3.8}$$

Now, given that the circles are touching, we know that:

$$\frac{O_1T}{O_2T} = \frac{r_1}{r_2} \tag{1.2.3.9}$$

$$\frac{O_1T}{O_1O_2} = \frac{r_1}{r_1 + r_2} \tag{1.2.3.10}$$

Hence, using the equation of line of centers obtained, we have:

$$\mathbf{T} = \mathbf{O}_1 + \frac{r_1}{r_1 + r_2}(\mathbf{O}_2 - \mathbf{O}_1) \tag{1.2.3.11}$$

codes/Circle.py

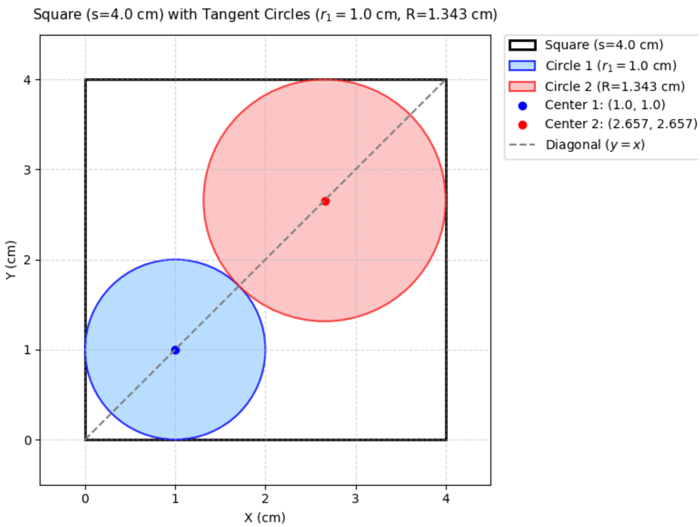

Fig. 1.2.3.2 Figure Constructed

1.2.4 Consider the contour C shown in the figure below. For the vector $\mathbf{F} = (x + 2y)\,\hat{e}_x + (2x + 4y)\,\hat{e}_y$, the integral $\oint_C \mathbf{F} \cdot d\mathbf{l} =$ ____________. Here $d\mathbf{l}$ represents an infinitesimal length along the contour C. (GATE AE 2026)

a) 0
b) 2
c) 4
d) 6

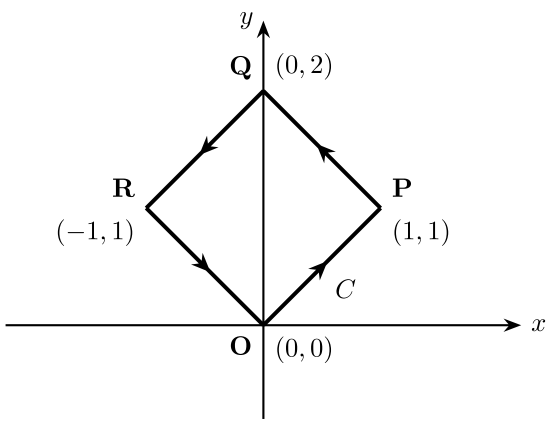

Fig. 1.2.4.1 Contour C

Solution:
We can break the closed path C into its four straight line paths: $O \to P \to Q \to R \to O$.

$$\mathbf{F} \cdot d\mathbf{l} = \begin{pmatrix} x+2y \\ 2x+4y \end{pmatrix} \cdot \begin{pmatrix} dx \\ dy \end{pmatrix} \tag{1.2.4.1}$$

$$\mathbf{F}^T d\mathbf{l} = \begin{pmatrix} x+2y \\ 2x+4y \end{pmatrix}^T \begin{pmatrix} dx \\ dy \end{pmatrix} = \begin{pmatrix} x+2y & 2x+4y \end{pmatrix} \begin{pmatrix} dx \\ dy \end{pmatrix} \tag{1.2.4.2}$$

$$= \big((x+2y)dx + (2x+4y)dy\big) \tag{1.2.4.3}$$

The general line integral expression to evaluate along any path is:

$$\mathbf{I} = \int (x+2y)\ dx + (2x+4y)\ dy \tag{1.2.4.4}$$

Path $O\,(0,0) \to P\,(1,1)$

- Equation of line: $y = x \implies dy = dx$
- Limits: x goes from 0 to 1

$$\mathbf{I_1} = \int_0^1 (x+2x)\ dx + (2x+4x)\ dx = \int_0^1 9x\,dx = \left[\frac{9x^2}{2}\right]_0^1 = \frac{9}{2} \tag{1.2.4.5}$$

Path $P\,(1,1) \to Q\,(0,2)$

- Equation of line: $y = 2-x \implies dy = -dx$
- Limits: x goes from 1 to 0

$$\mathbf{I_2} = \int_1^0 (x+2\,(2-x))\ dx + (2x+4\,(2-x))\,(-dx) \tag{1.2.4.6}$$

$$\mathbf{I_2} = \int_1^0 [(4-x)-(8-2x)]\ dx = \int_1^0 (x-4)\ dx = \left[\frac{x^2}{2} - 4x\right]_1^0 = \frac{7}{2} \tag{1.2.4.7}$$

Path $Q\,(0,2) \to R\,(-1,1)$

- Equation of line: $y = x+2 \implies dy = dx$
- Limits: x goes from 0 to -1

$$\mathbf{I_3} = \int_0^{-1} (x+2\,(x+2))\ dx + (2x+4\,(x+2))\ dx = \int_0^{-1} (9x+12)\ dx \tag{1.2.4.8}$$

$$\mathbf{I_3} = \left[\frac{9x^2}{2} + 12x\right]_0^{-1} = \left(\frac{9}{2} - 12\right) - 0 = -\frac{15}{2} \tag{1.2.4.9}$$

Path $R\,(-1,1) \to O\,(0,0)$

- Equation of line: $y = -x \implies dy = -dx$
- Limits: x goes from -1 to 0

$$\mathbf{I_4} = \int_{-1}^{0} (x + 2(-x))\, dx + (2x + 4(-x))(-dx) = \int_{-1}^{0} x\, dx \tag{1.2.4.10}$$

$$\mathbf{I_4} = \left[\frac{x^2}{2}\right]_{-1}^{0} = 0 - \frac{1}{2} = -\frac{1}{2} \tag{1.2.4.11}$$

Summing the integrals across all four line segments:

$$\mathbf{I} = \mathbf{I_1} + \mathbf{I_2} + \mathbf{I_3} + \mathbf{I_4} \tag{1.2.4.12}$$

$$\mathbf{I} = \frac{9}{2} + \frac{7}{2} - \frac{15}{2} - \frac{1}{2} = \frac{16 - 16}{2} = 0 \tag{1.2.4.13}$$

codes/Line_Integral.py

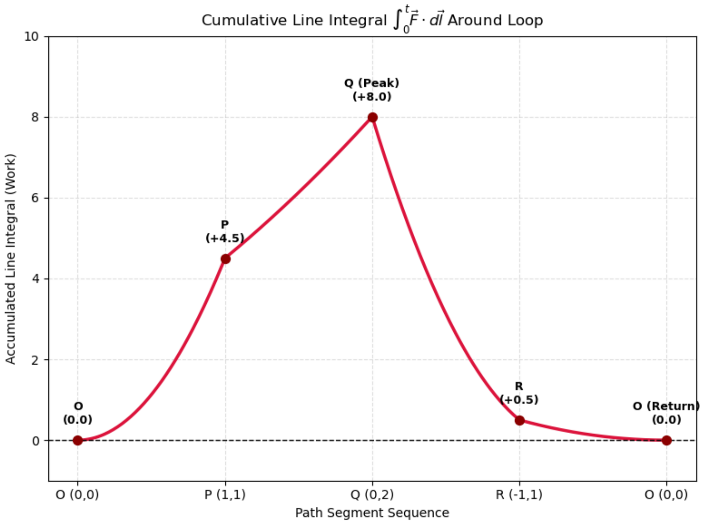

Fig. 1.2.4.2 Visualising the Cancellation of Integrals

1.2.5 The Fourier series representation of a square wave is shown in the figure below. The fluctuations seen near $x = \pm 1$ are named after which one of the following scientists? (GATE AE 2026)

a) Cauchy
b) Fourier
c) Gibbs
d) Laplace

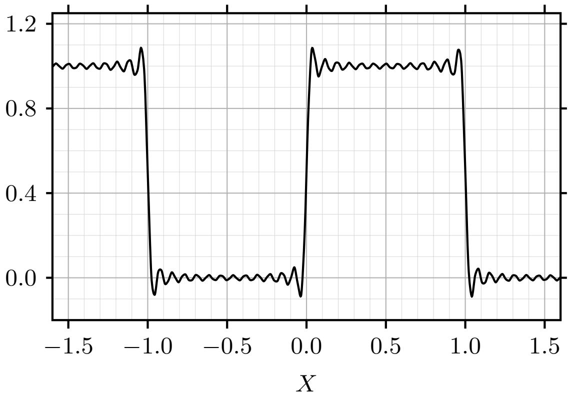

Fig. 1.2.5.1 Fourier Square Wave Representation

Solution:
Every periodic function can be expressed as a sum of sines and cosines, as seen from equation (1.1.4.1). Gibbs phenomenon is the overshoot and high-frequency oscillation (”ringing”) that occurs when approximating a function with jump discontinuities using a finite (truncated) Fourier series. Even as the number of terms N approaches infinity, the overshoot near the jump discontinuity does not vanish; instead, it narrows in width while maintaining a fixed percentage height relative to the size of the jump.

Reason: When approximating a step function or square wave, Fourier series synthesize the shape using smooth, continuous sinusoids ($\sin$ and $\cos$). Because continuous functions cannot instantaneously step from one value to another, the Fourier series overcompensates near the vertical edge.

1.2.6 If a matrix can be written as $\mathbf{A} = \mathbf{u}\mathbf{v}^{\mathbf{T}}$, where both u and v are n-dimensional real-valued non-zero column vectors, then the rank of the matrix $\mathbf{A}$ is ______. (GATE AE 2026)

Solution:
The matrix $\mathbf{A}$ is formed by taking the outer product of two $n \times 1$ non-zero column vectors $\mathbf{u}$ and $\mathbf{v}$:

$$\mathbf{A} = \mathbf{u}\mathbf{v}^{\mathbf{T}} \tag{1.2.6.1}$$

Writing $\mathbf{u}$ and $\mathbf{v}^{\mathbf{T}}$ explicitly:

$$\mathbf{u} = \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{pmatrix}, \quad \mathbf{v}^{\mathbf{T}} = \begin{pmatrix} v_1 & v_2 & \dots & v_n \end{pmatrix} \tag{1.2.6.2}$$

Multiplying $\mathbf{u}$ and $\mathbf{v^T}$ gives the $n \times n$ matrix $\mathbf{A}$:

$$\mathbf{A} = \begin{pmatrix} u_1v_1 & u_1v_2 & \dots & u_1v_n \\ u_2v_1 & u_2v_2 & \dots & u_2v_n \\ \vdots & \vdots & \ddots & \vdots \\ u_nv_1 & u_nv_2 & \dots & u_nv_n \end{pmatrix} \tag{1.2.6.3}$$

Notice that each column of $\mathbf{A}$ can be written as a scalar multiple of vector $\mathbf{u}$:

- 1st column: $v_1\mathbf{u}$
- 2nd column: $v_2\mathbf{u}$
- j^{th} column: $v_j\mathbf{u}$

The rank of a matrix is equal to the number of non zero rows in its echeleon form. When we reduce the above matrix into the echeleon form, we notice that each row is a multiple of another as shown above. Therefore, when we apply elementary row transformations by multiplying the rows with the necessary constants and subtracting, all rows but one become non zero. Hence, the rank of the matrix becomes 1.

Example:

Consider 3-dimensional non-zero vectors $\mathbf{u} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} 1 \\ 4 \\ 2 \end{pmatrix}$.

The matrix $\mathbf{A}$ is formed as:

$$\mathbf{A} = \mathbf{u}\mathbf{v^T} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \begin{pmatrix} 1 & 4 & 2 \end{pmatrix} \tag{1.2.6.4}$$

$$= \begin{pmatrix} 1 & 4 & 2 \\ 2 & 8 & 4 \\ 3 & 12 & 6 \end{pmatrix} \tag{1.2.6.5}$$

Applying elementary row transformations step-by-step:

Eliminate the non-zero entry in the second row using $R_2 \to R_2 - 2R_1$:

$$\mathbf{A} \xrightarrow{R_2 \to R_2 - 2R_1} \begin{pmatrix} 1 & 4 & 2 \\ 0 & 0 & 0 \\ 3 & 12 & 6 \end{pmatrix} \tag{1.2.6.6}$$

Eliminate the non-zero entry in the third row using $R_3 \to R_3 - 3R_1$:

$$\xrightarrow{R_3 \to R_3 - 3R_1} \begin{pmatrix} 1 & 4 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \tag{1.2.6.7}$$

The matrix is now in row echelon form with only 1 non-zero row. Hence, rank $(\mathbf{A}) = 1$.

1.2.7 An $n \times n$ square matrix $\mathbf{A}$ satisfies $\mathbf{A^T} = \mathbf{A^{-1}}$. The determinant of this matrix may take which of the following value(s)? (GATE AE 2026)

a) 1
b) -1
c) n
d) 0

Solution:
We are given that matrix A satisfies:

$$\mathbf{A}^{\mathbf{T}} = \mathbf{A}^{-\mathbf{1}} \tag{1.2.7.1}$$

Taking the determinant of both sides of the given matrix equation:

$$\left|\mathbf{A}^{\mathbf{T}}\right| = \left|\mathbf{A}^{-\mathbf{1}}\right| \tag{1.2.7.2}$$

Using (1.1.5.1):

$$\left|\mathbf{A}^{\mathbf{T}}\right| = \left|\mathbf{A}\right| \tag{1.2.7.3}$$

$$\left|\mathbf{A}^{-\mathbf{1}}\right| = \frac{1}{\left|\mathbf{A}\right|} \tag{1.2.7.4}$$

Replacing both sides using the above properties gives:

$$\left|\mathbf{A}\right| = \frac{1}{\left|\mathbf{A}\right|} \tag{1.2.7.5}$$

Multiplying both sides by $\left|\mathbf{A}\right|$:

$$\left(\left|\mathbf{A}\right|\right)^2 = 1 \implies \left|\mathbf{A}\right| = \pm 1 \tag{1.2.7.6}$$

1.2.8 Given the velocity and position vectors of a satellite along with its gravitational parameter $\mu = GM$, determine the trajectory of the satellite. (GATE AE 2026)

a) Circle
b) Hyperbola
c) Parabola
d) Straight Line

Position vector:

$$\mathbf{r} = 8000\hat{p} + 9000\hat{q} \text{ km} \tag{1.2.8.1}$$

Velocity vector:

$$\mathbf{v} = -6\hat{p} + 6\hat{q} \text{ km/s} \tag{1.2.8.2}$$

Gravitational parameter:

$$\mu = 398600 \text{ km}^3/\text{s}^2 \tag{1.2.8.3}$$

Solution:

The magnitude of the position vector $\mathbf{r}$ is:

$$r = |\mathbf{r}| = \sqrt{8000^2 + 9000^2} = \sqrt{64 \times 10^6 + 81 \times 10^6} = \sqrt{145 \times 10^6} \approx 12041.59 \text{ km} \tag{1.2.8.4}$$

The square of the velocity magnitude v^2 is:

$$v^2 = |\mathbf{v}|^2 = (-6)^2 + (6)^2 = 36 + 36 = 72 \text{ (km/s)}^2 \tag{1.2.8.5}$$

The specific mechanical energy (energy per unit mass) is given by:

$$E = \text{Kinetic Energy per unit mass} + \text{Potential Energy per unit mass} \tag{1.2.8.6}$$

$$E = \frac{1}{2}v^2 - \frac{\mu}{r} \tag{1.2.8.7}$$

Substituting the calculated values:

$$E = \frac{72}{2} - \frac{398600}{12041.59} \tag{1.2.8.8}$$

$$= +2.898 \text{ km}^2/\text{s}^2 \tag{1.2.8.9}$$

Any conic section in a 2D plane (p, q) is expressed by the general equation:

$$S(p, q) = ap^2 + 2hpq + bq^2 + 2gp + 2fq + c = 0 \tag{1.2.8.10}$$

For a satellite trajectory defined in polar coordinates by:

$$r = \frac{p_{\text{orb}}}{1 + e\cos\theta} \tag{1.2.8.11}$$

where $p = r\cos\theta$ and $q = r\sin\theta$:

$$r + ep = p_{\text{orb}} \implies r^2 = (p_{\text{orb}} - ep)^2 \tag{1.2.8.12}$$

Substituting $r^2 = p^2 + q^2$:

$$p^2 + q^2 = p_{\text{orb}}^2 - 2ep_{\text{orb}}p + e^2p^2 \tag{1.2.8.13}$$

Rearranging into standard form $S(p, q) = 0$:

$$\left(1 - e^2\right)p^2 + 0 \cdot pq + 1 \cdot q^2 + 2\left(ep_{\text{orb}}\right)p + 0 \cdot q - p_{\text{orb}}^2 = 0 \tag{1.2.8.14}$$

Comparing coefficients yields:

$$a = 1 - e^2, \quad b = 1, \quad c = -p_{\text{orb}}^2, \quad h = 0, \quad g = ep_{\text{orb}}, \quad f = 0 \tag{1.2.8.15}$$

The full 3D homogeneous quadratic form discriminant matrix M is defined as:

$$M = \begin{pmatrix} a & h & g \\ h & b & f \\ g & f & c \end{pmatrix} = \begin{pmatrix} 1 - e^2 & 0 & ep_{\text{orb}} \\ 0 & 1 & 0 \\ ep_{\text{orb}} & 0 & -p_{\text{orb}}^2 \end{pmatrix} \tag{1.2.8.16}$$

Expanding $|M|$ along the second row:

$$\Delta = |M| = 1 \cdot \begin{vmatrix} 1-e^2 & ep_{\text{orb}} \\ ep_{\text{orb}} & -p_{\text{orb}}^2 \end{vmatrix} = -p_{\text{orb}}^2 \left(1-e^2\right) - e^2 p_{\text{orb}}^2 = -p_{\text{orb}}^2 \tag{1.2.8.17}$$

Since $p_{\text{orb}} \neq 0$, $\Delta \neq 0$. The curve is strictly a non-degenerate conic.

Conic Family Test ($|A| = ab - h^2$):
The principal 2×2 submatrix $A = \begin{pmatrix} a & h \\ h & b \end{pmatrix}$ gives:

$$|A| = \left(1-e^2\right)(1) - 0^2 = 1 - e^2 \tag{1.2.8.18}$$

Specific Angular Momentum (h_0):

$$h_0 = |r \times v| = |8000\,(6) - 9000\,(-6)\,| = 102000 \text{ km}^2/\text{s} \tag{1.2.8.19}$$

Eccentricity Squared (e^2):

$$e^2 = 1 + \frac{2\varepsilon h_0^2}{\mu^2} = 1 + \frac{2\,(2.898)\,(102000)^2}{(398600)^2} \approx 1 + 0.3795 = 1.3795 \tag{1.2.8.20}$$

$$|A| = 1 - e^2 = 1 - 1.3795 = -0.3795 \tag{1.2.8.21}$$

From (1.1.5.11), we can say that the trajectory is a Hyperbola

codes/Satellite.py

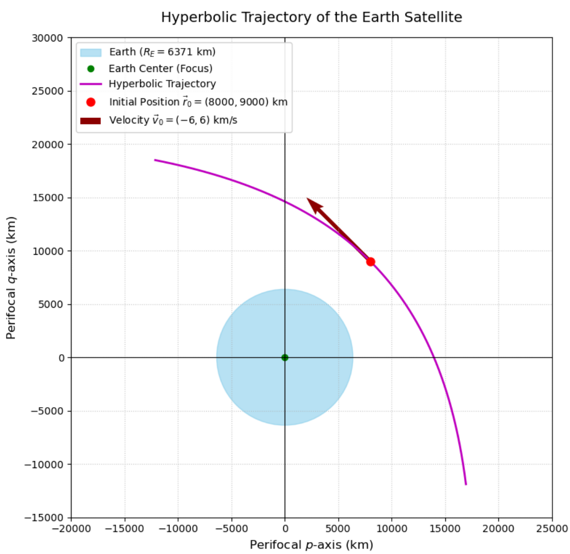

Fig. 1.2.8.1 Trajectory of the Satellite

1.2.9 For the matrix $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, if the relation $a + b = c + d$ holds and $a, b, c, d \neq 0$, then which one of the following statements about A is FALSE? (GATE AE 2026)

a) $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ is an eigenvector
b) $\lambda = a + b$ is an eigenvalue
c) $\lambda = d - b$ is an eigenvalue
d) $\lambda = d + b$ is an eigenvalue

Solution:

Given that $a + b = c + d = k$:

$$\begin{pmatrix} a & b \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = k = \begin{pmatrix} c & d \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \tag{1.2.9.1}$$

$$\implies \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} k \\ k \end{pmatrix} \tag{1.2.9.2}$$

$$\implies \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = k \begin{pmatrix} 1 \\ 1 \end{pmatrix} \tag{1.2.9.3}$$

$\implies \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ is an eigenvector and $k = a + b$ is an eigenvalue.

From the given relation $a + b = c + d$:

$$a - c = d - b = -(b - d) = k' \tag{1.2.9.4}$$

Evaluating the matrix-vector multiplication:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} a - b \\ c - d \end{pmatrix} = \begin{pmatrix} k' \\ -k' \end{pmatrix} = k' \begin{pmatrix} 1 \\ -1 \end{pmatrix} \tag{1.2.9.5}$$

$\implies \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ is an eigenvector and $d - b$ is the 2^{nd} eigenvalue.

codes/Eigenvectors.py

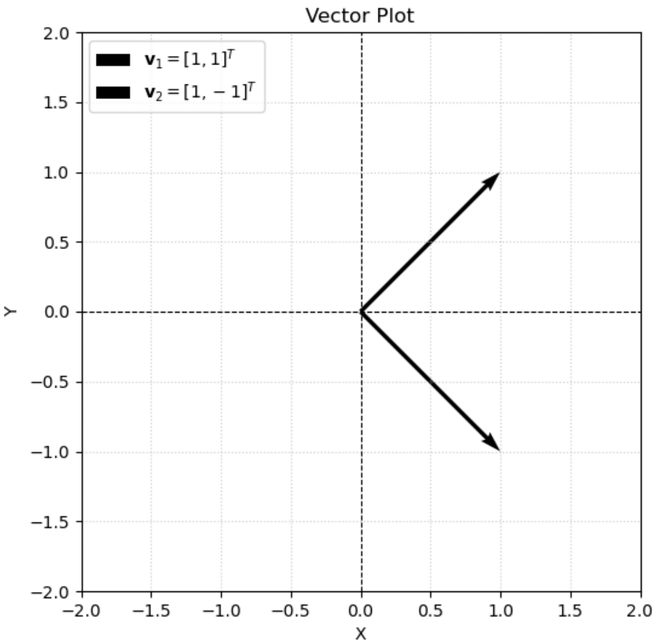

Fig. 1.2.9.1 Eigen vectors plotted

1.2.10 Consider the differential equation:

$$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = 0 \tag{1.2.10.1}$$

with initial conditions $y|_{x=0} = 0$ and $\left.\frac{dy}{dx}\right|_{x=0} = 1$. Find the value of the slope $\frac{dy}{dx}$ at $x = \ln(2)$ rounded off to three decimal places. (GATE AE 2026)

a) **Solution:**

Let $Y(s) = \mathcal{L}\{y(x)\}$. Taking the Laplace transform on both sides:

$$\mathcal{L}\left\{\frac{d^2y}{dx^2}\right\} + 2\mathcal{L}\left\{\frac{dy}{dx}\right\} + \mathcal{L}\{y\} = 0 \tag{1.2.10.2}$$

From equations (1.1.2.4) and (1.1.2.8):

$$\left[s^2Y(s) - sy(0) - y'(0)\right] + 2\left[sY(s) - y(0)\right] + Y(s) = 0 \tag{1.2.10.3}$$

Substituting the given initial conditions $y\left(0\right)=0$ and $y'\left(0\right)=1$:

$$(s^2Y(s)-1)+2sY(s)+Y(s)=0 \tag{1.2.10.4}$$

$$(s^2+2s+1)Y(s)=1 \tag{1.2.10.5}$$

$$Y(s)=\frac{1}{(s+1)^2} \tag{1.2.10.6}$$

Taking the inverse Laplace transform by applying the first frequency shift theorem((1.1.2.9)) on (1.1.2.17) gives the solution $y\left(x\right)$:

$$y\left(x\right)=\mathcal{L}^{-1}\left\{\frac{1}{(s+1)^2}\right\}=xe^{-x} \tag{1.2.10.7}$$

Differentiating $y\left(x\right)$ with respect to x:

$$\frac{dy}{dx}=\frac{d}{dx}\left(xe^{-x}\right)=\left(1-x\right)e^{-x} \tag{1.2.10.8}$$

Substituting $x=\ln\left(2\right)$:

$$\left.\frac{dy}{dx}\right|_{x=\ln(2)}=\left(1-\ln\left(2\right)\right)e^{-\ln(2)} \tag{1.2.10.9}$$

Since $e^{-\ln(2)}=\frac{1}{2}$:

$$\left.\frac{dy}{dx}\right|_{x=\ln(2)}=\frac{1-\ln\left(2\right)}{2}\approx\frac{1-0.693147}{2}=0.153426 \tag{1.2.10.10}$$

Rounding off to three decimal places yields: 0.153

b) Solution with Euler's Method:
Substituting equations (1.1.3.5) and (1.1.3.8) into our differential equation, we obtain:

$$\frac{y_{n+2}-2y_{n+1}+y_n}{h^2}+2\left(\frac{y_{n+1}-y_n}{h}\right)+y_n=0 \tag{1.2.10.11}$$

Rearranging, we get:

$$y_{n+2}=2(1-h)y_{n+1}-(1-h)^2y_n \tag{1.2.10.12}$$

Our initial conditions are:

$$y_0=y\left(0\right)=0, y_1=y_0+h(y'\left(0\right)) \tag{1.2.10.13}$$

Assuming $ln2=0.7$ and $h=0.01$ and evaluating the forward difference at $n=70$, we get:

$$y'_{70}=0.145 \tag{1.2.10.14}$$

This yields $\frac{dy}{dx}$ at $x = ln2$ as: 0.145

codes/Euler.py

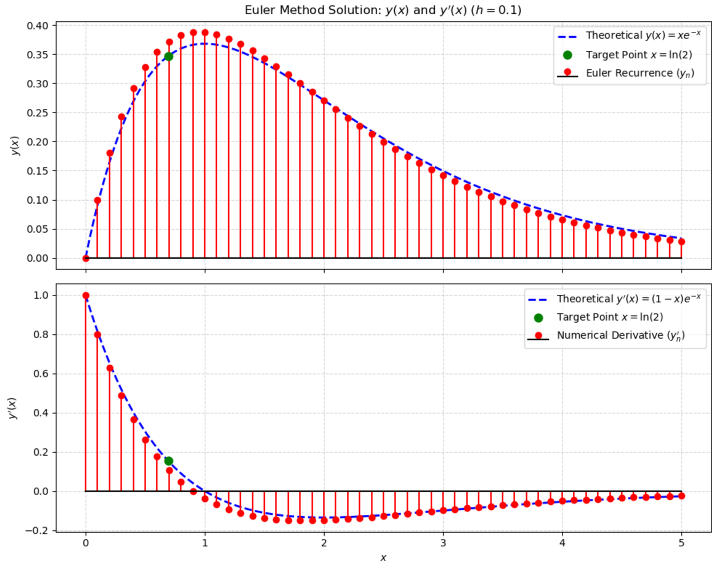

Fig. 1.2.10.1 Theoretical Plot vs Plot obtained with Euler's Forward Difference Method

1.2.11 Find the minimum value of the function $f(x) = |x| + |2x+3|$ for real x. (GATE AE 2026)

Solution:

The critical points occur where the expressions inside the absolute values equal zero:

$$x = 0 \tag{1.2.11.1}$$

$$2x + 3 = 0 \implies x = -\frac{3}{2} = -1.5 \tag{1.2.11.2}$$

These two critical points divide the real line into three sub-intervals:

$$(-\infty, -1.5), \quad [-1.5, 0), \quad \text{and} \quad [0, \infty) \tag{1.2.11.3}$$

- For $x < -1.5$:
  Both expressions inside the absolute values are negative, so $|x| = -x$ and $|2x+3| = -(2x+3)$:

$$f(x) = -x - (2x+3) = -3x - 3 \tag{1.2.11.4}$$

- For $-1.5 \leq x < 0$:

The term x is negative ($|x| = -x$) while $2x{+}3$ is non-negative ($|2x+3| = 2x{+}3$):

$$f(x) = -x + (2x+3) = x+3 \tag{1.2.11.5}$$

- For $x \geq 0$:
  Both expressions are non-negative, so $|x| = x$ and $|2x+3| = 2x+3$:

$$f(x) = x + (2x+3) = 3x+3 \tag{1.2.11.6}$$

Combining these cases gives the complete piecewise definition:

$$f(x) = \begin{cases} -3x-3, & \text{if } x < -1.5 \\ x+3, & \text{if } -1.5 \leq x < 0 \\ 3x+3, & \text{if } x \geq 0 \end{cases} \tag{1.2.11.7}$$

Analyzing the behavior of each linear segment:

- On $(-\infty, -1.5)$, the line $f(x) = -3x-3$ has a negative slope (-3), so $f(x)$ strictly decreases as x approaches -1.5.
- On $[-1.5, 0)$, the line $f(x) = x+3$ has a positive slope $(+1)$, so $f(x)$ strictly increases as x moves away from -1.5.
- On $[0, \infty)$, the line $f(x) = 3x+3$ has a positive slope $(+3)$, so $f(x)$ continues to increase.

Since $f(x)$ decreases for $x < -1.5$ and increases for $x > -1.5$, the minimum value occurs at $x = -1.5$:

$$f(-1.5) = -1.5 + 3 = 1.5 \tag{1.2.11.8}$$

codes/Modulus_Function.py
codes/Modulus_cvxpy.py

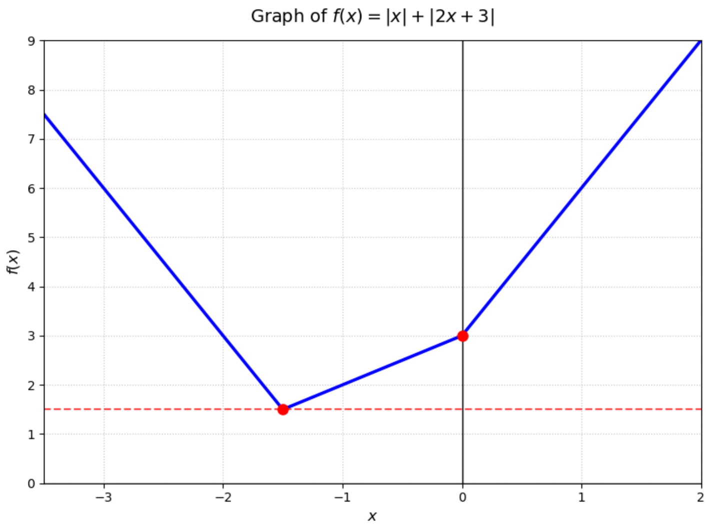

Fig. 1.2.11.1 Plot of the function

1.2.12 A bar, spring, and mass system is arranged in series.

- Bar: $E = 200$ GPa, $A = 100$ mm^2, $L = 100$ mm
- Spring Stiffness: $k = 200$ kN/mm
- Mass: $M = 100$ kg

Find the natural frequency of free vibration ω_n in rad/s. Also solve the differential equation obtained to obtain the position, velocity and accelaration of the system. (GATE AE 2026)

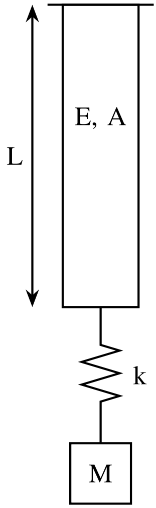

Fig. 1.2.12.1 Diagram of the system

Solution:

The axial stiffness of the uniform bar is:

$$k_{\text{bar}} = \frac{AE}{L} \tag{1.2.12.1}$$

Converting parameters to standard SI units:

$$E = 200 \times 10^9 \text{ N/m}^2 \tag{1.2.12.2}$$

$$A = 100 \times 10^{-6} \text{ m}^2 = 10^{-4} \text{ m}^2 \tag{1.2.12.3}$$

$$L = 100 \times 10^{-3} \text{ m} = 0.1 \text{ m} \tag{1.2.12.4}$$

$$k_{\text{bar}} = \frac{10^{-4} \times 200 \times 10^9}{0.1} = 2 \times 10^8 \text{ N/m} \tag{1.2.12.5}$$

The bar and the helical spring are connected in series. Therefore:

$$\frac{1}{k_{\text{eq}}} = \frac{1}{k_{\text{bar}}} + \frac{1}{k} \tag{1.2.12.6}$$

With $k = 200$ kN/mm $= 2 \times 10^8$ N/m:

$$k_{\text{eq}} = \frac{k_{\text{bar}} \cdot k}{k_{\text{bar}} + k} = \frac{\left(2 \times 10^8\right)\left(2 \times 10^8\right)}{2 \times 10^8 + 2 \times 10^8} = 10^8 \text{ N/m} \tag{1.2.12.7}$$

The natural frequency of free vibration is:

$$\omega_n = \sqrt{\frac{k_{\text{eq}}}{M}} = \sqrt{\frac{10^8}{100}} = \sqrt{10^6} = 1000 \text{ rad/s} \tag{1.2.12.8}$$

When the mass M is at rest, let the system stretch by δ_{st}

$$\sum F = 0 \implies Mg - k_{\text{eq}}\delta_{\text{st}} = 0 \implies Mg = k_{\text{eq}}\delta_{\text{st}} \tag{1.2.12.9}$$

When the mass is displaced downward by $x(t)$, applying Newton's second law ($\sum F = M\ddot{x}$):

$$M\frac{d^2x}{dt^2} = Mg - k_{\text{eq}}(\delta_{\text{st}} + x) \tag{1.2.12.10}$$

$$M\frac{d^2x}{dt^2} = Mg - k_{\text{eq}}\delta_{\text{st}} - k_{\text{eq}}x \tag{1.2.12.11}$$

Substituting the static equilibrium condition $Mg = k_{\text{eq}}\delta_{\text{st}}$:

$$M\frac{d^2x}{dt^2} = (k_{\text{eq}}\delta_{\text{st}}) - k_{\text{eq}}\delta_{\text{st}} - k_{\text{eq}}x \tag{1.2.12.12}$$

$$M\frac{d^2x}{dt^2} = -k_{\text{eq}}x \tag{1.2.12.13}$$

$$\frac{d^2x}{dt^2} = \frac{-k_{\text{eq}}}{M}x = -w_n^2 x \tag{1.2.12.14}$$

Let the initial displacement of the system be x_0 and inital velocity be v_0. From result (1.1.2.32), we get:

$$\mathcal{L}^{-1}\left(X(s)\right) = x(t) = x_0\cos(w_n t) + \frac{v_0}{w_n}\sin(w_n t) \tag{1.2.12.15}$$

Substituting the value of w_n,

$$\mathcal{L}^{-1}\left(X(s)\right) = x(t) = x_0\cos(1000t) + \frac{v_0}{1000}\sin(1000t) \tag{1.2.12.16}$$

codes/Spring.py

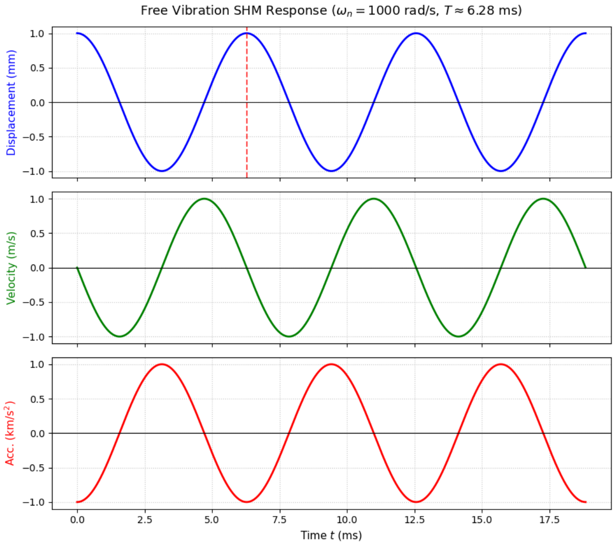

Fig. 1.2.12.2 Accelaration, Velocity and Position of the System